

Generation Interconnection System Impact Study Report

For Queue Project Q22-2125

Hilltop Alte 138 kV

August 22, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2125. The Interconnection Customer proposes a 240.9 MW solar photovoltaic facility interconnecting at the Hilltop Alte 138 kV substation in the New Jersey area. The request was received on June 25, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 677 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 7 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$2,823,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The MAGNOLIA 1708 - WHITE ROCK-NIKE 138 KV Line is overloaded to 114.1% of its 668 MVA emergency rating for the loss of the CRAWFORDSVILLE POWER PLANT - PINE TREE ACRES SUBSTATION 138 KV. The project contributes 19.2% DFAX (46.2 MW) to this overload.

JAN PHYL - BABINE LAKE 138 KV L: 112.0% loading (591 MVA) following the outage of the MAGOLIA 1708 - WEST ORANGE 138 KV line; DFAX = 15.0%.

SNE - SANSUM N ELEC - JAN PHYL 138 KV: 110.0% loading (662 MVA) following the outage of the W ORANGE - MAGNOLIA 1708 138 KV; DFAX = 15.0%.

SNE - SANSUM N ELEC - MULTITRADE OF PTTSYLVANIA LP 138 KV: 109.1% loading (701 MVA) following the outage of the W ORANGE - MAGNOLIA 1708 138 KV; DFAX = 15.0%.

The DAM HTS - ARTESIAN 138 KV is overloaded to 104.3% of its 302 MVA emergency rating for the loss of the MAGNOLIA 1708 - W ORANGE 138 KV. The project contributes 6.0% DFAX (14.6 MW) to this overload.

The W ALDWELL - NORTHEASTERN 138 KV is overloaded to 103.1% of its 251 MVA emergency rating for the loss of the MAGNOLIA 1708 - W ORANGE 138 KV L. The project contributes 6.6% DFAX (15.8 MW) to this overload.

SUNTREE - ORANGE 138 KV: 102.2% loading (308 MVA) following the outage of the WEST ORANGE - MAGNOLIA 1708 138 KV; DFAX = 6.6%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
MAGNOLIA 1708 - WHITE ROCK-NIKE 138 KV Line	Rebuild line with higher-capacity conductor	\$1,805,000	15% (\$268,000)
JAN PHYL - BABINE LAKE 138 KV	Rebuild line with higher-capacity conductor	\$868,000	23% (\$200,000)
SNE - SANSUM N ELEC - JAN PHYL 138 KV	Rebuild line with higher-capacity conductor	\$834,000	23% (\$192,000)
MULTITRADE OF PITTSYLVANIA LP - SNE - SANSUM N ELEC SUBSTATION 138 KV	Reconductor line	\$347,000	23% (\$80,000)
DAM HTS - ARTESIAN 138 KV	Reconductor line	\$465,000	14% (\$66,000)
W ALDWELL - NORTHEASTERN 138 KV	Reconductor line	\$1,462,000	100% (\$1,462,000)
SUNTREE - ORANGE 138 KV	Reconductor line	\$555,000	100% (\$555,000)
	Total allocated to project		\$2,823,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.